HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Final Mock Examination — Set B (Lectures 1–12)

Comprehensive coverage of ISLP Chapters 2–8, 10 and 13,
with emphasis on Chapters 7 (splines/GAMs), 8 (trees), 10 (deep learning) and 13 (multiple testing)

— Version with detailed solutions —

Duration:	**120 minutes**
Total credit:	**120 points**
Allowed aids:	non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: ______________________ Matriculation no.: ______________

Problem	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**Total**
Maximum points	16	8	20	16	20	20	20	**120**
Achieved points								

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- All numerical constants you need (powers of e, logarithms, $(0.95)^{10}$, t-quantiles) are provided in the problems.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Moving beyond linearity — splines (16 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Throughout this problem, consider regression splines built on a single predictor X using $K = 5$ interior knots.

(a) **[4 P]** A *cubic spline* is a piecewise cubic polynomial that is continuous and has continuous first and second derivatives at every knot. Derive how many degrees of freedom (free parameters, counting the intercept) such a spline uses for a general number K of interior knots, by counting the parameters of the polynomial pieces and subtracting the constraints imposed at the knots. Then evaluate the count for $K = 5$.

(b) **[4 P]** How many degrees of freedom does a *natural* cubic spline with the same $K = 5$ knots use? State the additional constraint a natural spline imposes, in which region of the predictor space it acts, and why practitioners find this constraint useful.

(c) **[4 P]** A *smoothing spline* is the function g that minimises

$$\sum_{i=1}^{n}\big(y_i - g(x_i)\big)^2 + \lambda \int g''(t)^2\, dt.$$

Explain the meaning of the two terms, and describe the shape of the solution $\hat{g}$ in the two limiting cases $\lambda \to 0$ and $\lambda \to \infty$, giving the effective degrees of freedom that $\hat{g}$ approaches in each case.

(d) **[4 P]** Write down the *truncated-power basis* representation of a cubic spline with a single knot at ξ, define the truncated-power basis function $(x-\xi)^3_+$ explicitly, and explain in one sentence why adding this term keeps the fit continuous with continuous first and second derivatives at ξ.

Solution

(a) Degrees of freedom of a cubic spline. With K interior knots the range of X is divided into $K+1$ intervals, and on each interval the fit is a cubic with 4 coefficients, giving $4(K+1) = 4K+4$ free parameters. At every knot the requirement that g, g' and g'' join continuously imposes 3 constraints, i.e. $3K$ constraints in total. Hence

$$\mathrm{df} = 4(K+1) - 3K = K + 4.$$

For $K = 5$: $\mathrm{df} = 5 + 4 = \mathbf{9}$ (equivalently the basis intercept, x, x^2, x^3 plus one truncated-power term per knot, $4 + K$).

(b) Natural cubic spline. A natural spline adds the requirement that the function be *linear beyond the two boundary knots*, i.e. on the outer regions $X < \xi_1$ and $X > \xi_K$. Forcing the quadratic and cubic terms to vanish in each of those two regions removes 2 parameters per side, 4 in total:

$$\mathrm{df} = (K+4) - 4 = K, \qquad K = 5: \ \mathrm{df} = \mathbf{5}.$$

Usefulness: ordinary polynomials and unconstrained splines behave erratically near the boundaries, where the data are sparse; forcing linearity there reduces the *variance* of predictions at and beyond the outer knots, at the cost of only a little extra bias.

(c) Smoothing spline. The first term (the RSS) rewards closeness to the observed data, while the second term penalises *roughness*: g'' measures the local curvature of g, so $\int g''(t)^2 dt$ is large for wiggly functions, and $\lambda \geq 0$ controls the trade-off. *$\lambda \to 0$:* the penalty vanishes, so $\hat{g}$ may be arbitrarily wiggly and *interpolates* every training point (zero training RSS, extreme overfitting); the effective degrees of freedom approach n. *$\lambda \to \infty$:* any curvature becomes infinitely costly, forcing $g'' \equiv 0$, so $\hat{g}$ is a straight line, namely the ordinary *least-squares regression line* (effective $\mathrm{df} \to 2$). For intermediate λ (chosen e.g. by LOOCV) the effective df decrease smoothly from n down to 2.

(d) Truncated-power basis, single knot ξ.

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x-\xi)^3_+, \qquad (x-\xi)^3_+ = \begin{cases} (x-\xi)^3, & x > \xi \\ 0, & x \leq \xi \end{cases}$$

At $x = \xi$ the added function $(x-\xi)^3_+$ and its first and second derivatives are all 0, so including it alters only the *third* derivative across the knot; the fit therefore stays continuous with continuous first and second derivatives — precisely the smoothness that defines a cubic spline.

Problem 2: Generalised additive models (8 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

(a) **[3 P]** Write down the general form of a *generalised additive model* (GAM) for a quantitative response Y and predictors $X_1, \ldots, X_p$, and explain in one sentence how it generalises multiple linear regression.

(b) **[3 P]** State one important *interpretability advantage* of GAMs (concerning the fitted functions f_j) and their central structural *limitation*, including how that limitation can be repaired.

(c) **[2 P]** Describe in one or two sentences how the *backfitting* algorithm fits a GAM.

Solution

(a) Model form.

$$Y = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p) + \varepsilon.$$

Multiple linear regression is the special case in which every $f_j(X_j) = \beta_j X_j$; a GAM replaces each straight-line term by a flexible smooth function f_j (e.g. a natural or smoothing spline, or a local regression) while retaining the *additive* structure.

(b) Advantage and limitation. *Advantage (interpretability):* because the effects add up, the contribution of each predictor is carried by its own function f_j, which can be plotted and read off one panel at a time with the other predictors held fixed — non-linear effects stay as easy to inspect as coefficients in a linear model. *Limitation:* pure additivity rules out *interactions* — the effect of X_j is not allowed to depend on the value of another X_k. Repair: introduce interaction terms by hand, e.g. a product $X_j \times X_k$ or a low-dimensional smooth surface $f_{jk}(X_j, X_k)$ as an extra additive component.

(c) Backfitting. Loop over the predictors and repeatedly re-estimate one f_j at a time (with a smoother) using the *partial residuals* $r_i = y_i - \beta_0 - \sum_{k\neq j} f_k(x_{ik})$, keeping the other fitted functions fixed; repeat the cycles until the f_j no longer change appreciably.

Problem 3: Regression and classification trees by hand (20 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

A logistics firm records, for $n = 6$ regional depots, the number of delivery vans x and the number of parcels delivered per day y (in thousands):

Depot i	1	2	3	4	5	6
Vans x_i	10	12	14	16	18	20
Parcels y_i	6	4	5	11	13	12

A regression tree with a *single* split at cutpoint s partitions the depots into $R_1 = \{i : x_i < s\}$ and $R_2 = \{i : x_i \geq s\}$ and predicts the region mean $\hat{y}_{R_m}$ in each region.

(a) **[8 P]** For each of the two candidate cutpoints $s = 15$ and $s = 17$, determine the two regions, their means, and the total residual sum of squares $\text{RSS}(s) = \sum_{m=1}^{2} \sum_{i\in R_m} (y_i - \hat{y}_{R_m})^2$. Which cutpoint does recursive binary splitting prefer, and why?

(b) **[2 P]** Using the preferred split, predict the number of parcels for a new depot with $x = 13$ vans.

(c) **[6 P]** A *classification* tree for a different task has a terminal node containing 12 training observations from three classes: 6 “Premium”, 3 “Standard” and 3 “Basic”. Compute the class proportions, the Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ and the entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ of this node (*use* $\ln 0.5 = -0.6931$ *and* $\ln 0.25 = -1.3863$). Which class does the node predict?

(d) **[4 P]** In *cost-complexity pruning* one minimises, over subtrees $T \subseteq T_0$ of the large tree T_0,

$$\sum_{m=1}^{|T|} \sum_{i:\, x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha\, |T|.$$

Explain the role of α: what happens for $\alpha = 0$, what happens as α grows, and how is α chosen in practice?

Solution

(a) Evaluating the two cutpoints. For reference the root mean is $\bar{y} = (6+4+5+11+13+12)/6 = 51/6 = 8.5$.

Cutpoint $s = 15$*:* $R_1 = \{1,2,3\}$ with y-values $\{6,4,5\}$, mean $\hat{y}_{R_1} = 15/3 = 5$; $R_2 = \{4,5,6\}$ with y-values $\{11,13,12\}$, mean $\hat{y}_{R_2} = 36/3 = 12$.

$$\begin{aligned}\mathrm{RSS}_{R_1} &= (6-5)^2 + (4-5)^2 + (5-5)^2 = 1+1+0 = 2,\\ \mathrm{RSS}_{R_2} &= (11-12)^2 + (13-12)^2 + (12-12)^2 = 1+1+0 = 2,\end{aligned}$$

so $\mathrm{RSS}(15) = 2+2 = \mathbf{4}$.

Cutpoint $s = 17$*:* $R_1 = \{1,2,3,4\}$ with y-values $\{6,4,5,11\}$, mean $\hat{y}_{R_1} = 26/4 = 6.5$; $R_2 = \{5,6\}$ with y-values $\{13,12\}$, mean $\hat{y}_{R_2} = 12.5$.

$$\mathrm{RSS}_{R_1} = (6-6.5)^2 + (4-6.5)^2 + (5-6.5)^2 + (11-6.5)^2 = 0.25 + 6.25 + 2.25 + 20.25 = 29,$$

$$\mathrm{RSS}_{R_2} = (13-12.5)^2 + (12-12.5)^2 = 0.25 + 0.25 = 0.5,$$

so $\mathrm{RSS}(17) = 29 + 0.5 = \mathbf{29.5}$.

Recursive binary splitting is *greedy* and takes the cutpoint with the smaller RSS: since $4 < 29.5$, it prefers $\mathbf{s = 15}$, which cleanly separates the three low-throughput depots (mean 5) from the three high-throughput depots (mean 12).

(b) Prediction for $x = 13$. $x = 13 < 15$, so the new depot falls into R_1 and the tree predicts $\hat{y} = \hat{y}_{R_1} = \mathbf{5}$ (i.e. 5,000 parcels per day).

(c) Node impurity (three classes). $\hat{p}_{\text{Prem}} = 6/12 = 0.5$, $\hat{p}_{\text{Std}} = 3/12 = 0.25$, $\hat{p}_{\text{Basic}} = 3/12 = 0.25$.

$$G = 0.5(1-0.5) + 0.25(1-0.25) + 0.25(1-0.25) = 0.25 + 0.1875 + 0.1875 = \mathbf{0.625}.$$

$$\begin{aligned}D &= -\big(0.5\ln 0.5 + 0.25\ln 0.25 + 0.25\ln 0.25\big)\\ &= -\big(0.5(-0.6931) + 0.5(-1.3863)\big) = 0.3466 + 0.6931 = \mathbf{1.040}.\end{aligned}$$

The node predicts the majority class, **Premium** (estimated probability 0.5). Both G and D would be 0 for a pure node; here the node is quite impure, consistent with the split carrying three classes.

(d) Role of α. $\alpha \geq 0$ is the price charged per terminal node, trading the subtree's fit (first term) against its size $|T|$. For $\alpha = 0$ the criterion reduces to the training RSS, so the full unpruned tree T_0 is optimal. As α increases, each extra leaf must "earn its keep"; a branch whose RSS reduction is smaller than α is pruned away, so the optimal subtree shrinks and one obtains a *nested sequence* of subtrees as α climbs (weakest-link pruning). In practice α is selected by *cross-validation* — pick the α giving the lowest CV error, then refit that subtree on the full data — which balances bias and variance rather than merely minimising training error.

Problem 4: Ensembles — bagging, random forests, boosting (16 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

(a) **[6 P]** Let $\hat{f}_1(x), \ldots, \hat{f}_B(x)$ be B identically distributed (but not independent) tree predictions at a fixed point x, each with variance σ^2 and pairwise correlation ρ. The variance of the bagged average is

$$\mathrm{Var}\Big(\tfrac{1}{B}\sum_{b=1}^{B}\hat f_b(x)\Big) = \rho\,\sigma^2 + \frac{1-\rho}{B}\,\sigma^2.$$

(i) Explain briefly why this formula shows that bagging reduces variance and which term limits the reduction. (ii) Evaluate the variance for $B = 500$, $\rho = 0.7$ and $\sigma^2 = 4$, and compare it with the variance of a single tree. (iii) What value does the variance approach as $B \to \infty$?

(b) **[4 P]** How do *random forests* modify bagging at each split, and what is the recommended subset size m for *classification* as a function of p (evaluate for $p = 36$)? Using the formula in (a), explain why this modification improves on bagging.

(c) **[4 P]** Name the three tuning parameters of *boosting* for regression trees and say in one sentence each what they control. What is the "slow learning" idea behind boosting?

(d) **[2 P]** What does the *out-of-bag* (OOB) error of a bagged model estimate, and why is it essentially free to obtain?

Solution

(a) Variance of the bagged average. (i) The second term $\frac{1-\rho}{B}\sigma^2$ shrinks to 0 as B grows, so averaging over many trees removes the "independent part" of their variability — this is the variance reduction of bagging. The first term $\rho\sigma^2$ does *not* depend on B: the bootstrap trees are grown on overlapping resamples of the same data (and share the same strong predictors), so they are positively correlated, and this common part of the variance cannot be averaged away. It is the floor that limits the reduction. (ii) For $B = 500$, $\rho = 0.7$, $\sigma^2 = 4$:

$$\mathrm{Var} = 0.7\cdot 4 + \frac{1-0.7}{500}\cdot 4 = 2.8 + \frac{0.3\cdot 4}{500} = 2.8 + 0.0024 = \mathbf{2.802},$$

versus $\sigma^2 = 4$ for a single tree — a reduction of about 30%. (iii) As $B \to \infty$ the variance approaches the floor $\rho\sigma^2 = 0.7 \cdot 4 = \mathbf{2.8}$. Enlarging B further does not overfit; it only stabilises the average.

(b) Random forests. At each split the tree may consider only a fresh *random subset* of m out of the p predictors as candidates. For classification the usual choice is $m \approx \sqrt{p}$, so for $p = 36$: $m = \sqrt{36} = \mathbf{6}$. Why it helps: in plain bagging a dominant predictor is chosen at the top split of nearly every tree, making the trees look alike and pushing ρ up. Withholding that predictor from about $(p-m)/p$ of the split searches *decorrelates* the trees, i.e. lowers ρ; by (a) this lowers the irreducible floor $\rho\sigma^2$ of the ensemble variance, which plain averaging can never reach.

(c) Boosting.

- *Number of trees B:* how many small trees are added in sequence; unlike bagging, boosting *can* overfit if B is too large, so B is tuned by cross-validation.
- *Shrinkage* λ (learning rate, e.g. 0.01 or 0.001): scales the contribution of each new tree and hence the speed of learning; a smaller λ usually requires a larger B.
- *Number of splits d per tree* (interaction depth): the complexity of each tree and the interaction order captured; often even $d = 1$ (stumps, an additive model) suffices.

Slow learning: each new tree is fit to the current *residuals*, and only a small fraction λ of it is added to the model, so the fit improves gradually and precisely where it is still weak, rather than fitting the data hard in a single step — which tends to generalise better.

(d) OOB error. Each bootstrap sample omits, on average, about one third of the observations ("out-of-bag"). Predicting each observation only from the trees that did *not* train on it, and aggregating, gives the OOB error — a valid estimate of the bagged model's *test* error. It is essentially free because it reuses a by-product of the bootstrap; no separate validation set or cross-validation loop is needed.

Problem 5: Neural networks by hand (20 points)

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Consider a small feed-forward network for a quantitative response with input $x = (x_1, x_2) = (2, 1)$, one hidden layer of two ReLU units ($g(z) = \max\{0, z\}$) and a linear output unit:

$$A_1 = g\big(-2 + 3\,x_1 - 1\,x_2\big), \qquad A_2 = g\big(1 - 2\,x_1 + 1\,x_2\big), \qquad f(x) = -1 + 2\,A_1 + 3\,A_2.$$

(a) **[8 P]** Carry out the forward pass step by step: compute both pre-activations, both activations A_1, A_2 and the output $f(x)$. Comment briefly on what the ReLU did to the second unit and why non-linear activations are essential.

(b) **[6 P]** A classification network with $K = 3$ classes returns the logits (output-layer pre-activations) $z = (z_1, z_2, z_3) = (2.0,\ 1.0,\ -1.0)$ for one observation whose true class is class 1. Compute all three softmax probabilities and the cross-entropy loss $-\ln \hat{p}_{\text{true}}$. *(Use* $e^2 = 7.389$*,* $e^1 = 2.718$*,* $e^{-1} = 0.368$ *and* $\ln 10.475 = 2.349$*.)*

(c) **[4 P]** Count the parameters of a fully connected network with $p = 6$ inputs, one hidden layer of $k = 4$ units and an output layer of $K = 3$ units, where every hidden and output unit carries a bias. Show the count separately for the two weight layers.

(d) **[2 P]** A convolutional layer receives a 32×32 image and applies filters of size $F = 5$ with padding $P = 0$ and stride $S = 1$. Compute the spatial size of the output feature map from $\big(W - F + 2P\big)/S + 1$.

Solution

(a) Forward pass. Pre-activations:

$$z_1 = -2 + 3 \cdot 2 - 1 \cdot 1 = -2 + 6 - 1 = 3, \qquad z_2 = 1 - 2 \cdot 2 + 1 \cdot 1 = 1 - 4 + 1 = -2.$$

ReLU activations:

$$A_1 = g(3) = \max\{0, 3\} = 3, \qquad A_2 = g(-2) = \max\{0, -2\} = 0.$$

Output:

$$f(x) = -1 + 2 \cdot 3 + 3 \cdot 0 = -1 + 6 + 0 = \mathbf{5}.$$

The ReLU *switched the second unit off*: its negative pre-activation is clipped to 0, so unit 2 contributes nothing for this particular input (though it could for others). This thresholding is exactly the source of non-linearity: with a purely linear g, a network of any depth would collapse to a single linear function of x.

(b) Softmax and cross-entropy. Exponentials $e^{2.0} = 7.389$, $e^{1.0} = 2.718$, $e^{-1.0} = 0.368$ sum to $7.389 + 2.718 + 0.368 = 10.475$. Softmax probabilities:

$$\hat{p}_1 = \frac{7.389}{10.475} = 0.705, \qquad \hat{p}_2 = \frac{2.718}{10.475} = 0.259, \qquad \hat{p}_3 = \frac{0.368}{10.475} = 0.035,$$

which sum to $0.999 \approx 1$. Cross-entropy loss for the true class 1:

$$-\ln \hat{p}_1 = -\ln \frac{e^2}{10.475} = -\big(2 - \ln 10.475\big) = 2.349 - 2 = \mathbf{0.349}.$$

(Equivalently $-\ln 0.705 \approx 0.349$; a perfect prediction $\hat{p}_1 = 1$ would give loss 0.)

(c) Parameter count for $6 \to 4 \to 3$. Input $\to$ hidden: $6 \cdot 4 = 24$ weights $+\,4$ biases $= 28$. Hidden $\to$ output: $4 \cdot 3 = 12$ weights $+\,3$ biases $= 15$. Total: $28 + 15 = \mathbf{43}$ parameters.

(d) Convolution output size.

$$\frac{W - F + 2P}{S} + 1 = \frac{32 - 5 + 2 \cdot 0}{1} + 1 = \frac{27}{1} + 1 = 28,$$

so the output feature map is $\mathbf{28} \times \mathbf{28}$. With no padding a 5×5 filter at stride 1 shrinks each spatial dimension by $F - 1 = 4$.

Problem 6: Multiple testing (20 points)

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

(a) **[4 P]** A lab runs $m = 10$ *independent* hypothesis tests, each at level $\alpha = 0.05$, and suppose all 10 null hypotheses are true. Define the family-wise error rate (FWER) and compute it here *(use* $(0.95)^{10} = 0.5987$*)*. Interpret the result in one sentence.

(b) **[7 P]** A team screens $m = 6$ candidate biomarkers; the ordered p-values are

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.009	0.012	0.015	0.040	0.300

Control the FWER at $\alpha = 0.05$ (i) with *Bonferroni* and (ii) with the *Holm step-down* procedure. Give the relevant thresholds, state exactly which hypotheses each method rejects, and explain why Holm can never reject fewer hypotheses than Bonferroni.

(c) **[6 P]** Apply the *Benjamini–Hochberg* (BH) procedure at FDR level $q = 0.05$ to the same six p-values: give the thresholds jq/m, determine which hypotheses are rejected, and state precisely what "FDR control at $q = 0.05$" guarantees for the resulting set of rejections.

(d) **[3 P]** A pharmaceutical company runs a *confirmatory* Phase-III trial with $m = 4$ pre-specified primary endpoints; a single significant endpoint could support regulatory approval of the drug. Should the company control the FWER or the FDR? Justify your recommendation.

Solution

(a) FWER for $m = 10$ independent tests. FWER $=$ Pr(at least one Type I error) $=$ $\Pr(V \geq 1)$, where V counts the true nulls that are falsely rejected. With independence and all nulls true,

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{10} = 1 - (0.95)^{10} = 1 - 0.5987 = \mathbf{0.4013}.$$

Interpretation: although each individual test uses the 5% level, there is about a 40% chance of at least one false discovery across the ten tests — testing many hypotheses without adjustment makes spurious findings very likely.

(b) Bonferroni and Holm at $\alpha = 0.05$. *Bonferroni:* reject $H_{(j)}$ iff $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$. Only $p_{(1)} = 0.002 \leq 0.00833$ ✓; already $p_{(2)} = 0.009 > 0.00833$ ×. $\Rightarrow$ Bonferroni rejects $H_{(1)}$ alone: **1 rejection**. (Note $p_{(2)} = 0.009$ is below the naive 0.05 cutoff, yet Bonferroni does not reject it.)
Holm: compare $p_{(j)}$ with $\alpha/(m - j + 1)$ in order and stop at the first failure:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.009	0.012	0.015	0.040	0.300
$\alpha/(m-j+1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
reject?	✓	✓	✓	✓	stop	—

$0.002 \le 0.00833$, $0.009 \le 0.0100$, $0.012 \le 0.0125$, $0.015 \le 0.0167$, but $0.040 > 0.0250$: stop. $\Rightarrow$ Holm rejects $H_{(1)}, H_{(2)}, H_{(3)}, H_{(4)}$: **4 rejections**.
Why Holm $\supseteq$ Bonferroni: Holm's thresholds $\alpha/(m-j+1)$ are all $\ge \alpha/m$ (with equality only at $j=1$), so any p-value that clears the strict Bonferroni bar also clears Holm's more lenient later hurdles — Holm is uniformly at least as powerful, while still controlling the FWER at α with no independence assumption.

(c) Benjamini–Hochberg at $q = 0.05$. Thresholds $jq/m = j \cdot 0.05/6$:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.009	0.012	0.015	0.040	0.300
jq/m	0.00833	0.0167	0.0250	0.0333	0.0417	0.0500
$p_{(j)} \le jq/m$?	yes	yes	yes	yes	**yes**	no

L = largest j with $p_{(j)} \le jq/m$: here $j = 5$ (since $0.040 \le 0.0417$, while $0.300 > 0.0500$). BH rejects *all* $H_{(j)}$ with $j \le L$, i.e. $H_{(1)}, \ldots, H_{(5)}$: **5 rejections**. Guarantee: FDR $= E\big[V/\max(R,1)\big] \le q = 0.05$ — *on average* at most 5% of the rejected hypotheses are false discoveries. It does *not* promise that at most 5% of these particular 5 rejections are false, nor does it bound the probability of making any false rejection (that would be FWER control).

(d) Confirmatory trial recommendation. Control the **FWER**. Here even a *single* false positive is very costly — it could lead to approving an ineffective drug — so the study must guard against any Type I error, not merely keep their average proportion small. With only $m = 4$ pre-specified endpoints, an FWER procedure (e.g. Holm or Bonferroni at $0.05/4 = 0.0125$) sacrifices little power. FDR control, appropriate for large exploratory screens where a few false leads are tolerable, is too permissive for a confirmatory, decision-driving trial.

Problem 7: Cumulative essentials (Chapters 2–6) (20 points)

Lecture slides: Chapters 2–6 (Lectures 1–8).

(a) **[4 P]** A medical screening classifier is evaluated on 200 patients, giving the confusion matrix

	Predicted diseased	Predicted healthy
Actually diseased	40	10
Actually healthy	30	120

Compute the overall error rate, the sensitivity, the specificity and the precision (positive predictive value). Which single number would trace out the ROC curve if it were varied, and in which direction would sensitivity and specificity move?

(b) **[4 P]** Consider K-nearest-neighbours. (i) Using the training set (x, y): $(1,3)$, $(4,6)$, $(6,8)$, $(7,7)$, $(10,12)$, predict the response at $x_0 = 5$ with $k = 3$ (KNN regression, unweighted average). (ii) Explain how k governs the bias–variance trade-off, which extreme ($k = 1$ or k large) is the most flexible, and why the *training* error of 1-NN is always 0.

(c) **[4 P]** Explain in one or two sentences what k-fold cross-validation does, then give one reason to prefer $k = 5$ or 10 over *LOOCV* and one reason to prefer it over a *single* validation split.

(d) **[2 P]** You have $p = 2{,}000$ predictors but expect only a handful to be truly related to the response. Ridge or lasso — which penalty is more appropriate, and where would principal-components regression (PCR) sit relative to your choice?

(e) **[6 P]** Match each scenario to exactly one method from {*linear regression*, *lasso*, *logistic regression*, *random forest*} (each method used exactly once) and justify each choice in one sentence:

(i) Estimate how much one additional year of schooling adds to annual earnings, with a confidence interval, from $n = 5{,}000$ survey responses.
(ii) Predict a continuous blood-pressure outcome from $p = 8{,}000$ genetic markers for $n = 150$ patients, and report a short list of the relevant markers.
(iii) Classify each incoming e-mail as spam or not, and explain to auditors how each feature changes the odds of an e-mail being spam.
(iv) Predict machine failure from hundreds of mixed sensor variables with strong non-linear effects and interactions; only predictive accuracy matters.

Solution

(a) Confusion-matrix metrics. Here $TP = 40$, $FN = 10$, $FP = 30$, $TN = 120$, total 200; 50 patients are truly diseased and 150 truly healthy.

$$\text{error rate} = \frac{FP + FN}{200} = \frac{30 + 10}{200} = \mathbf{0.200}, \qquad \text{sensitivity} = \frac{TP}{TP + FN} = \frac{40}{50} = \mathbf{0.800},$$

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{120}{150} = \mathbf{0.800}, \qquad \text{precision} = \frac{TP}{TP + FP} = \frac{40}{70} = \mathbf{0.571}.$$

Varying the *classification threshold* on the predicted probability traces out the ROC curve. Lowering the threshold labels more patients "diseased", which *raises* sensitivity (fewer missed cases) but *lowers* specificity (more false alarms); raising it does the reverse — the ROC curve captures exactly this trade-off.

(b) K-nearest-neighbours. (i) Distances from $x_0 = 5$ are $|1-5| = 4$, $|4-5| = 1$, $|6-5| = 1$, $|7-5| = 2$, $|10-5| = 5$; the three nearest neighbours are $x = 4, 6, 7$ with responses $6, 8, 7$, so $\hat{y}(5) = (6+8+7)/3 = 21/3 = \mathbf{7}$. (ii) Small k gives a very flexible fit: low bias but high variance, since the prediction hangs on a few nearby points and a wiggly boundary. Large k averages over many neighbours, giving a smooth, stable fit: high bias but low variance. The most flexible extreme is $\mathbf{k = 1}$ (flexibility $\propto 1/k$). The training error of 1-NN is always 0 because each training point's single nearest neighbour is *itself*, so it predicts its own observed response exactly — which is why training error is no guide to test performance.

(c) k-fold cross-validation. It splits the data into k roughly equal folds; each fold in turn serves as the validation set while the model is trained on the other $k-1$ folds, and the k resulting errors are averaged. *vs. LOOCV:* the n LOOCV fits use almost identical training sets, so their errors are highly correlated and their average has *higher variance* than 5- or 10-fold CV (and LOOCV costs n fits, absent shortcut formulas). *vs. a single validation split:* a single split gives a high-variance estimate that depends heavily on the particular partition and is biased upward because only part of the data trains the model; k-fold CV averages over k folds and trains on $(k-1)/k$ of the data, giving a more stable, less biased estimate.

(d) Lasso. The ℓ_1 penalty drives many coefficients *exactly to zero* and thus performs variable selection — ideal when the truth is sparse (a few real signals among 2,000 predictors) and an interpretable small model is wanted. Ridge only shrinks coefficients and keeps all 2,000 predictors, so it tends to lose here. PCR is likewise *not* sparse: it regresses on a few principal components, so all original variables load on the retained components and no variable is truly dropped — like ridge it does not perform selection, so it does not deliver the short interpretable list that lasso does.

(e) Matching.

- (i) **Linear regression**: a quantitative response with $n \gg p$ and the goal of *inference* — an effect size with a confidence interval — exactly what OLS with standard errors provides.
- (ii) **Lasso**: $p \gg n$ makes OLS infeasible; the ℓ_1 penalty regularises the fit and returns a short list of relevant markers.
- (iii) **Logistic regression**: a binary response with interpretable coefficients ($e^{\hat{\beta}_j}$ are odds ratios) — well suited to explaining odds to auditors.
- (iv) **Random forest**: automatically captures non-linearities and interactions, is accurate and robust with many mixed predictors, and interpretability is not required.

End of the examination — good luck!